Evidence for Evolution



Comparative Studies in Biochemistry

DNA

Overview

- Idea of evolution and evidence for it accepted by vast majority of scientists
- Has been controversial for reasons unrelated to the scientific evidence
- Important to look at evidence to support scientific theory
- Recent advances in technology provide further evidence to support the theory of evolution



Comparative Studies in Biochemistry - DNA

- DNA
  - uses sequences of 4 nitrogenous bases
  - Codes for protein production
  - All living things use the same 4 bases, the same DNA code
  - Sequence of bases varies
    - Mutation
    - Natural selection
    - Genetic Drift
  - As living things reproduce, code differs over generations between offspring, despite common ancestor, but some stays the same.
    - More distantly related: greater variation
    - Closely related: more shared

Table 15.1 Relationship between humans and great apes using DNA differences

Primates being compared	DNA difference (%)
Human – chimpanzee	1.2
Chimpanzee – gorilla	1.2
Human – gorilla	1.6
Chimpanzee – orang-utan	1.8
Human – orang-utan	2.4
Gorilla – orang-utan	2.4

Comparative Studies in Biochemistry - DNA

- Chromosomes also contain sequences of bases that don't appear to code for anything:
  - "junk DNA"
  - Also vary across generations
  - Can also be compared for similarities and differences to assess relatedness.
- Example: Endogenous retroviruses (ERV)
  - Viral sequence that has become part of the organism's genome in the germline cells
  - Passed on to each new generation
  - Make up 8% of human genome
  - Other primates have some of the same sequences in the same locations, indicating that we, and they, are descended from a common ancestor who was originally infected with the virus.

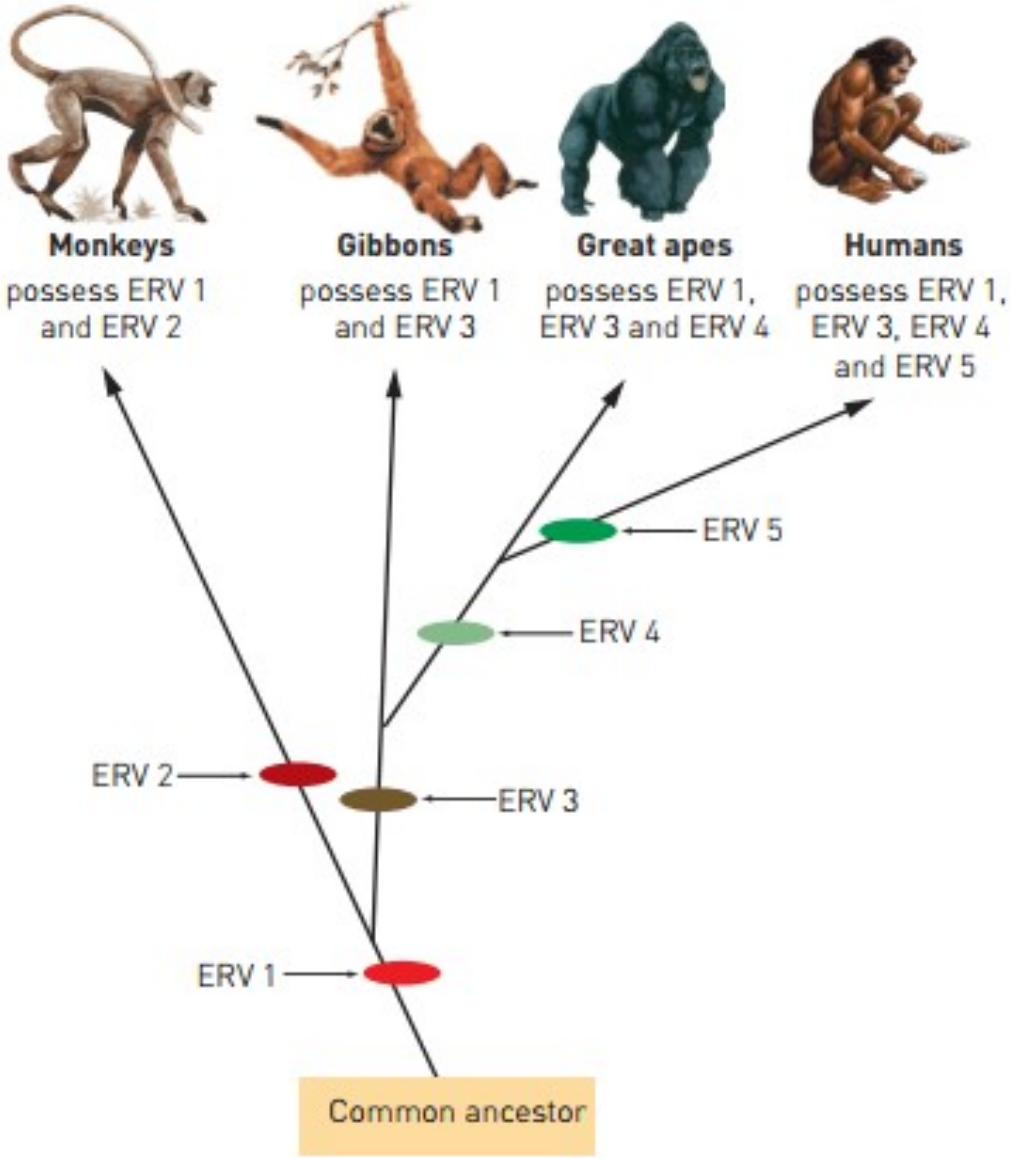


Figure 15.1 A simplified example of how endogenous retroviruses could be used to trace common ancestry: the great apes and humans have a more recent common ancestor as they share more endogenous retroviruses.

Comparative Studies in Biochemistry - DNA

- PCR can be used to amplify tiny amounts of DNA
  - Facilitates sequencing
  - Allows small amounts of DNA from fossil record to be analysed
- Electrophoresis
  - Also used to create profile and compare
- Bacterial enzymes (restriction enzymes)
  - Used to cut up DNA to sequence it.

** Processes are a key part of sequencing DNA for analysis of relatedness between species